

A Privacy-Preserving Pipeline for Local Digitization of Handwritten Persian Sleep-Laboratory Forms: Architecture, Reproducibility, and Operational Validation

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Abstract

Background: Retrospective digitization of paper clinical questionnaires is difficult when a document combines handwritten text, right-to-left script, checkbox grids, tables, and annotated diagrams. **Objective:** We describe and operationally validate a local, configuration-driven pipeline for digitizing a nine-page Persian sleep-laboratory questionnaire while avoiding transmission of form images to an external OCR service. **Methods:** The pipeline renders PDF pages, detects layout regions and table cells with PaddleOCR/PaddleX models, identifies page types from structural fingerprints, corrects under-detected tables, numbers cells in right-to-left order, recognizes each detected crop with the locally installed Bina 0.2 Rizeh Persian recognizer, detects tick marks and diagram annotations, and writes page-level JSON and research CSV outputs. A reproducibility test was performed on one nine-page PDF supplied with the implementation. Operational outcomes were extracted from pipeline artifacts and Bina’s model-reported recognition scores. **Results:** The test produced nine page database files, nine annotated review images, and 337 cell-level OCR records without an inference exception. All 337 records contained an output string; however, only 97/337 (28.8%) exceeded the configured recognition-score threshold of 0.50 and were therefore treated as readable by the pipeline. The mean, median, and range of the reported

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scores were 0.291, 0.172, and 0–0.989, respectively. The run produced a 78-column research CSV and completed the global-dataset export logic with duplicate prevention by record identifier. **Conclusions:** The implementation is suitable as a privacy-preserving research data-ingestion tool with explicit human-review triage, but the present evidence does not establish field-level accuracy, character error rate, clinical validity, or superiority over another OCR system. A multi-form reference-standard study is required before clinical or epidemiological claims are made.

Keywords: Persian handwriting recognition, clinical form digitization, document image analysis, OCR, privacy-preserving health informatics, reproducible software

1. Introduction

Paper questionnaires remain important sources of clinical and research data, particularly where electronic collection is unavailable or where historical archives predate electronic health records. Retrospective digitization is not equivalent to ordinary text recognition. A clinical form may contain printed instructions, free-form handwriting, tabular response options, checkbox marks, numeric measurements, and annotations on preprinted diagrams. These elements require different image-processing assumptions and different error controls.

Persian forms add a further challenge. Text is written in a right-to-left script with contextual glyph shaping, while numbers and Latin abbreviations may occur in left-to-right runs. An OCR system can therefore produce characters that appear visually plausible but are stored in the wrong logical order. Handwritten text recognition is also highly sensitive to writing style, segmentation, scan quality, and the presence of ruling lines. Recent surveys emphasize that handwriting recognition remains dependent on language, document type, writer variability, and the availability of representative annotated data [1].

The software studied here was originally designed around a cloud large-vision-language-model workflow. That design was subsequently changed to a local Bina 0.2 Rizeh recognizer, which is based on a PP-OCRv6 medium recognition graph and provides a wrapper that converts visual-order Arabic-script output to logical Persian order [2]. The current system also includes a

local browser portal, per-run artifacts, an append-only global research CSV, and a human-review signal based on recognition scores.

The purpose of this paper is therefore narrower and more reproducible than the earlier manuscript. We document the current implementation, report a directly reproducible operational test, and define what the test does and does not demonstrate. We do not describe the system as clinically validated. In accordance with contemporary reporting guidance for artificial-intelligence evaluation, claims about accuracy require a defined reference standard, representative test data, participant or document flow, and uncertainty estimates [3,4]. Those materials were not present in the submitted archive and are not inferred from model confidence scores.

1.1. Contributions

This paper makes four contributions. First, it documents a modular local pipeline for a mixed-content Persian clinical form. Second, it describes the integration constraints required to run the custom Bina export reproducibly, including explicit selection of `PP-OCrv6_medium_rec` and disabling CPU oneDNN/MKLDNN fusion for this graph. Third, it reports an artifact-level validation of the complete workflow on a nine-page PDF. Fourth, it distinguishes operational coverage and model confidence from true transcription accuracy and identifies the validation study required for publication as a clinical AI evaluation.

2. System and Implementation

2.1. Design requirements

The design was guided by four requirements. Form images should remain on the local computer during OCR. The system should preserve intermediate artifacts so that errors can be inspected. The structure of the questionnaire should be used to constrain extraction rather than asking one general-purpose model to infer the entire document. Finally, the research export should be stable across repeated runs and should support both per-form inspection and longitudinal accumulation of records.

2.2. Current processing workflow

The current implementation is executed by `pipeline/main.py`. The processing order is shown in Fig. 1 and consists of: PDF rendering and layout detection; table-cell detection; page identity recognition; image-mark

detection; table correction and secondary fallback; cell numbering; direct per-cell OCR; tick analysis; database finalization; annotated review generation; and local and global CSV export. The local portal presents eight user-facing progress stages, while table correction, fallback, cell numbering, and review generation are internal processing operations.

2.3. Layout, cell detection, and page identity

PDF pages are rendered with PyMuPDF and passed to PaddleOCR layout and table-cell detection components. The form-specific configuration stores expected structural fingerprints, including table and image counts and table-cell counts. A page recognizer maps source pages to semantic page numbers. Cell detection uses the wired-table cell detector and a primary confidence threshold configured in the project. Table correction compares observed structure with reference geometry and can invoke a secondary detection pass when a table appears under-detected. These operations are form-specific in their configuration even though their image-processing interfaces are modular.

2.4. Bina OCR integration

The original grouped-image Gemini path remains in the source tree as a legacy option, but the default provider is now Bina. Each detected cell crop is passed directly to the local recognizer. The implementation explicitly selects `PP-OCRv6_medium_rec`, because the generic PaddleOCR text-recognition constructor otherwise defaults to a different model name. On CPU, `oneDNN/MKLDNN` is disabled because the exported Bina graph produced an input-rank failure under the fused execution path. With this setting, the recognizer returned text and a model-reported score for the test crops.

The Bina wrapper converts Arabic-script visual order into logical order. The pipeline therefore stores the returned string directly and does not apply a second Arabic reshaping or bidirectional-display transformation. This distinction is essential for Persian research data: presentation-form glyphs and visual-order strings should not be confused with logical Unicode text. Scores below 0.50 are stored as `[unreadable]` in the research output, while the raw score remains available in the page-level OCR record.

Bina’s model card explicitly cautions that its confidence values are model scores rather than calibrated probabilities and that the bundled benchmark should not be interpreted as universal document accuracy [2]. Accordingly, this study uses the score threshold only as an operational review rule, not as an accuracy estimate.

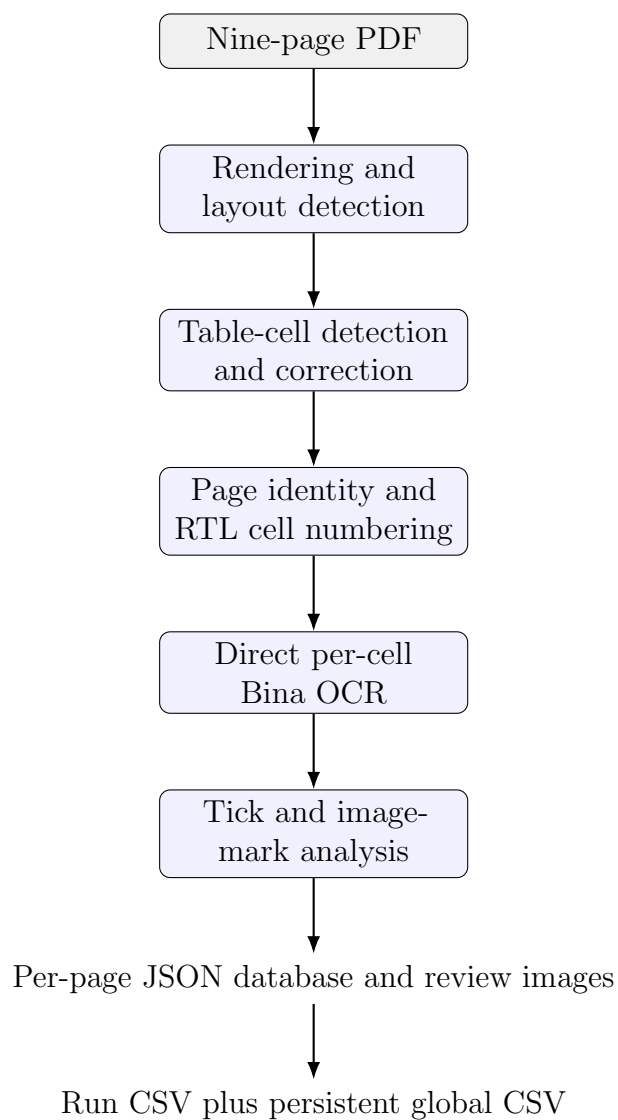


Figure 1: Current local processing architecture. The OCR stage receives individual cell crops rather than grouped composite images.

2.5. Tick marks and image annotations

The pipeline performs pixel-density-based tick analysis on checkbox and Likert-style response regions. It records row-level selections and reflects them into cell-level database records. Pages containing preprinted anatomical diagrams are compared with local reference images after alignment. Difference and binary-mark artifacts are retained for inspection. These components produce structured outputs but require independent reference-standard evaluation before their sensitivity or specificity can be claimed.

2.6. Data model and export

Each semantic page is saved as a JSON record containing tables, cells, OCR results, tick results, image-mark results, and timestamps. The OCR results are explicitly handed to the database builder before page serialization. A run-level research CSV is produced after all page records exist. The portal additionally supplies an absolute path to a persistent global CSV at the project root. Each completed record is identified by a stable record identifier; the exporter checks existing identifiers before appending, preventing duplicate rows when a completed job is retried.

3. Operational Validation

3.1. Validation question and data

The validation question was whether the current implementation could execute the complete local workflow on a supplied nine-page PDF and produce inspectable OCR, database, review-image, and research-export artifacts. The test was not designed to estimate transcription accuracy because no independently prepared, cell-level human reference transcription accompanied the PDF. No training, validation, or test partition was created from this single form.

The test PDF generated 337 cell crops. The Bina output for each crop contained a label, text string, score, source image, and result type. The pipeline threshold was 0.50. A record was classified as operationally readable when its score was at least 0.50; this classification is a triage rule, not a correctness label.

3.2. Reproducibility procedure

The test was executed in an isolated Python environment using Python 3.12 in the sandbox validation environment. The relevant runtime versions were PaddlePaddle 3.2.2, PaddleOCR 3.7.0, and PaddleX 3.7.2. The Windows launcher requests Python 3.11 because the project targets a validated Windows installation. The Bina model was installed separately from its public model repository and was not embedded in the source archive. The final pipeline was executed after the model and crop-generation stages had completed, and the focused OCR test was then repeated over all 337 crops.

The following outputs were checked: the number of OCR records; the number of nonempty OCR outputs; the distribution of model-reported scores; page-level JSON files; annotated review images; the run-level research CSV; and the duplicate-prevention behavior of the global exporter. The validation script is included with the manuscript materials.

3.3. Results

The focused Bina run returned 337/337 OCR records without an inference exception. All records contained an output string, but this included the explicit `[unreadable]` sentinel for low-score crops. Ninety-seven crops (28.8%) reached the configured score threshold and 240 (71.2%) were flagged for review. The mean score was 0.291, the median was 0.172, and the observed range was 0–0.989. The 10th and 90th percentiles were 0.000 and 0.806, respectively.

Table 1: Operational Bina results by semantic page for the supplied nine-page PDF.

Page	Crops	Score ≥ 0.50	Review	Mean score	Median score
1	16	10	6	0.645	0.646
2	8	0	8	0.186	0.224
3	100	20	80	0.204	0.000
4	50	10	40	0.185	0.000
5	54	13	41	0.239	0.000
6	30	23	7	0.668	0.836
7	52	12	40	0.224	0.000
8	10	3	7	0.424	0.457
9	17	6	11	0.452	0.418
Total	337	97	240	0.291	0.172

The complete pipeline subsequently produced nine page-level JSON files and nine annotated review images. The run-level CSV contained 78 columns and one record. The global-export regression test appended one record and retained one record after a repeated export of the same identifier, demonstrating idempotence for retry handling. These results establish execution and artifact integrity; they do not establish that the recognized strings or mark selections are correct.

4. Discussion

4.1. Interpretation

The current implementation provides a defensible software architecture for local research-data ingestion. Its most important practical property is not an unverified accuracy percentage but the separation of structural extraction, recognition, mark analysis, storage, and review. This separation makes it possible to inspect an uncertain crop, preserve the model score, and re-run or replace one stage without discarding the entire document.

The operational results also show why a confidence-aware human-review workflow is necessary. Although every crop produced a result record, only 28.8% passed the configured score threshold in the supplied PDF. The low threshold-pass rate was concentrated in pages containing dense response grids and mixed printed/handwritten content. Treating every nonempty output as correct would therefore overstate system performance. A production workflow should expose the score, original crop, and surrounding form context to a reviewer and should not silently convert low-confidence text into clinical data.

The local deployment model addresses an important governance concern. Form images and transcriptions remain within the local portal workspace during normal Bina processing. This is preferable to the former Gemini path when institutional policy prohibits external transmission. Local execution does not, however, by itself establish compliance with a particular legal or institutional framework. Retention, access control, encryption, and deletion policies remain deployment responsibilities.

4.2. Relation to prior work

Pipeline-based OCR systems have long been used to separate document handling, preprocessing, recognition, and postprocessing, allowing components to be exchanged and evaluated independently [5]. This design is especially appropriate for clinical forms, where field selection and reference-standard

construction are as important as the recognizer. Modern OCR toolkits expose separate document orientation, detection, and recognition modules [6], while handwriting-recognition surveys emphasize the effects of writer variation, segmentation, language, and dataset choice [1]. The current system follows this modular pattern but specializes the structural stages for a Persian sleep questionnaire and uses a local Persian recognizer for the cropped text regions.

4.3. Limitations

The central limitation is the absence of a human reference standard for the supplied test. Bina scores are not calibrated probabilities and cannot be converted into character accuracy, word accuracy, sensitivity, specificity, or clinical validity. The former manuscript’s claims of 91.4% field accuracy, 8.6% character error rate, a GPT-4o comparison, and a 15-fold speedup are not retained because the underlying annotated dataset, timing protocol, and comparator outputs were not included in the submitted archive and could not be independently reproduced.

The test used one PDF and therefore cannot estimate between-patient, between-writer, scanner, or form-version variability. The form configuration is also questionnaire-specific. Changing the questionnaire requires new structural fingerprints, expected cell counts, tick mappings, and review criteria. The local model is Persian-focused and is not evidence for other languages or document domains. The score threshold was selected as an operational rule and has not been calibrated against correctness.

The image-mark and tick modules were exercised as part of the full pipeline, but no independent annotations were available for estimating their accuracy. Similarly, successful creation of database and CSV files verifies software execution, not semantic validity. Finally, the current code retains a legacy Gemini branch even though Bina is the default. This increases flexibility but also increases maintenance and documentation burden; future releases should either isolate the legacy branch more clearly or remove it after migration is complete.

4.4. Required next study

A definitive accuracy study would sample forms across handwriting quality, scan quality, age, and form version. Two trained annotators would independently transcribe every target cell and mark every checkbox and diagram region, with adjudication of disagreements. Such a study would

report eligible and excluded forms, the reference-standard procedure, missing and illegible fields, writer and form characteristics, and document-level train/validation/test partitioning. It would include exact cell accuracy for categorical fields, character error rate and word error rate for free text, selection accuracy and agreement statistics for tick marks, localization metrics for diagram annotations, and confidence intervals clustered by form rather than treating cells as independent observations. Review workload at several score thresholds and an external audit of a held-out set would further characterize deployment suitability. These requirements follow the transparency principles of STARD and STARD-AI [3,4].

5. Conclusion

The updated system is a local, modular pipeline for converting a mixed-content Persian sleep-laboratory PDF into inspectable JSON and research CSV artifacts. Its current implementation uses direct per-cell Bina recognition, explicit logical-order Persian handling, confidence-based review flags, local database persistence, and an idempotent global dataset export. In a reproducibility test on one nine-page PDF, the pipeline processed 337 crops without an inference exception and generated all expected downstream artifacts. The same test showed that most crops were below the configured Bina score threshold, reinforcing the need for human review.

The appropriate current claim is therefore that the system is operationally reproducible and suitable for research-assisted digitization, not that it is clinically accurate or validated. A reference-standard evaluation on a representative multi-form cohort is the necessary next step before the system is used for automated clinical decisions or before comparative accuracy claims are made.

Ethics and data governance

This report concerns software engineering and operational reproducibility of a local document-processing pipeline. The validation used one de-identified archival PDF supplied for evaluation; no participants were recruited or contacted, no clinical intervention was performed, and no clinical decision was made. The PDF and all derived artifacts were processed locally. The institutional ethics determination and data-custodian authorization governing

the archival source should be reported according to the policy of the submitting institution; the present software test does not independently determine whether review, exemption, or waiver was required.

Data availability

The clinical PDF and derived OCR outputs are not publicly released because they originate from a clinical archive. The validation script, dependency versions, model identifier, and reported operational summaries are provided with the software materials. A synthetic or structurally equivalent test set may be released where permitted by the responsible institution.

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